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Notes on Classical Electron Dynamics: Lorentz–Dirac Equation

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ABSTRACT

1. THE LORENTZ EQUATION AND THE RUNAWAY PROBLEM

[Conventions, stated once here and used throughout: Gaussian units for electrodynamics, SI for mechanics, $c = 1$. Metric signature and the definition of u^α, a^α to be fixed here. Note that the 2006 text declared Gaussian units but evaluated τ_0 without the factor $1/4\pi\varepsilon_0$; the corrected value is used here from the start.]

1.1. *Electrodynamics does not supply an equation of motion for the charge*

Classical electrodynamics rests on three pillars: the conservation laws, chief among them the conservation of charge; Maxwell's equations; and the Lorentz force law. Maxwell's equations fix the field produced by a given distribution of charge and current. The Lorentz force law fixes the force exerted on a charge by a given field. Between them the two appear to account for everything an electrodynamical problem could ask for, and they do not: neither supplies the equation of motion of the charge itself.

The gap is structural rather than an oversight. To follow the motion of a charge one needs the force acting on it, hence the total field at its location, and the total field includes the field that the charge itself produces. Maxwell's equations say how that self-field is generated but not how it acts back; the Lorentz force law says how a field acts on a charge but is silent on whether the charge's own field belongs among the fields that act. Electrodynamics is complete as an account of how charges make fields and of how fields push charges, and it is incomplete exactly where the two meet on the same particle. Closing that gap is the subject of everything below.

1.2. *Three historical routes*

Any attempt to close the gap must begin by deciding what a charge is, and two pictures divide the subject between them. In the first the charge sits at a point. Its own field then diverges as the point is approached, and with the field the Lorentz force, so the force expression loses its meaning exactly where it is needed. One may object that the Lorentz force was only ever intended for external fields and that the self-field is simply not subject to it; but the objection buys nothing, because it leaves the self-force undetermined and the equation of motion no closer than before. That is the standing difficulty of the point picture. In the second the charge is spread over a finite region and treated as a continuous medium. Every quantity is then finite and the self-force is a well-defined integral over the distribution. The price is that like charges repel, so a body carrying charge of one sign must fly apart; bound charges nevertheless exist, and electrodynamics does not say what holds them together.

Three attempts, spread over thirty years, took these pictures in turn. Poincaré (1906) worked in the extended picture and supplied the missing agent by hand, adding to the electromagnetic stress–energy tensor a binding tensor — the Poincaré stresses — chosen so that the total is in equilibrium and the charged body holds. The construction does what is asked of it, but it postulates a force for the sole purpose of removing a single contradiction, with no other consequence and no independent evidence for its existence; in the century since, few have followed it.

Lorentz (1909) took a position between the two pictures: the electron is a sphere of finite radius carrying a uniform charge. This is concrete enough to yield a definite equation of motion, which is the subject of Sec. 1.3.

Dirac (1938) returned to the point picture, but gave up computing the self-force as the mutual action of one part of the charge on another. He imposed momentum–energy conservation on a thin tube surrounding the world line and read the equation of motion off the balance, so that the divergent self-field enters only through quantities that cancel. This is the route the rest of these notes follow; it is set out in Sec. 2.

These notes adopt the point picture, for the reasons that recommended it to Dirac. It is technically uniform; the model carries no internal structure that has to be specified in advance; the equation of motion comes out in a fixed form; and the construction generalizes, both to curved backgrounds and to the corresponding gravitational problem. The cost is that infinite quantities appear at intermediate stages, so that the force has to be extracted indirectly — either by balancing finite quantities, as in the conservation method, or by separating off a singular part and discarding it. In what sense those operations are legitimate is a question that runs through everything below.

1.3. *The Lorentz model and its three defects*

Lorentz's electron is a sphere of finite radius carrying a uniform charge: far enough into the extended picture that nothing diverges, structureless enough to remain tractable. What holds the charge together is left unanswered, which is precisely the question the Poincaré stresses were invented to settle. In motion the sphere is contracted into an ellipsoid.

Two forces then act on it. One is external. The other is a self-force, produced by the action of the charge elements on one another. At rest this self-force vanishes by symmetry. In motion it does not, and the reason is retardation: the field with which one element acts on a second is the field it carried at an earlier time, and the reaction of the second on the first is delayed in the same way, so the two no longer cancel. The charge elements taken by themselves therefore violate Newton's third law. Nothing is genuinely lost — the missing momentum resides in the field — but the failure is a fair measure of how far the problem has already moved away from mechanics.

1.3.1. *Deriving the self-force directly*

The calculation itself is not deep; the work is all in the expansion. Let the charge be distributed with density ρ over a rigid body of size R, moving slowly enough that magnetic contributions may be dropped to the order kept. The self-force is the double integral of the retarded interaction over all pairs of charge elements,

$$\mathbf{F}_{\text{self}} = \iint d^3r\, d^3r'\, \rho(\mathbf{r})\rho(\mathbf{r}')\, \mathbf{K}(\mathbf{r},\mathbf{r}';\, t-s)\,, \qquad s = \frac{|\mathbf{r}-\mathbf{r}'|}{c}, \tag{1}$$

in which the kernel $\mathbf{K}$ is evaluated not at the present time but one light-crossing time s earlier. Since $s \sim R/c$ is small, expand every retarded quantity about the present instant,

$$f(t-s) = f(t) - s\,\dot{f}(t) + \frac{s^2}{2}\ddot{f}(t) - \frac{s^3}{6}\dddot{f}(t) + \cdots\,, \tag{2}$$

and collect. Each additional power of s costs one factor of R/c and buys one more time derivative of $\mathbf{v}$, so the series organizes itself by derivative order: the term containing $d^n\mathbf{v}/dt^n$ carries R^{n-2}. The zeroth term is the static self-repulsion, which vanishes by symmetry for a rigid symmetric body, and what survives is

$$\mathbf{F}_{\rm self} = -\frac{4U_0}{3c^2}\,\dot{\mathbf{v}} \;+\; \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}} \;+\; \sum_{n\geq 3} \alpha_n \,\frac{e^2 R^{n-2}}{c^{n+1}}\frac{d^n\mathbf{v}}{dt^n}, \tag{3}$$

where U_0 is the electrostatic self-energy and the α_n are pure numbers fixed by the shape of the distribution.

The three groups of terms meet three quite different fates, set out in Table 1, and the whole of the subject descends from that split.

Term	Coefficient scales as	What becomes of it
$n = 1$, $\propto \dot{\mathbf{v}}$	R^{-1}, divergent	absorbed into the mass
$n = 2$, $\propto \ddot{\mathbf{v}}$	R^0, model-independent	survives as the radiation reaction
$n \geq 3$, $\propto d^n\mathbf{v}/dt^n$	R^{n-2}, vanishing	lost in the point limit

Table 1. The three groups of terms in the expansion Eq. (3), and what becomes of each as the radius of the body is taken to zero. Only the middle row survives, and the classical theory of the electron is the theory of that one row.

The first row is the divergence, and it is harmless only because of its form: being proportional to $\dot{\mathbf{v}}$, it cannot be distinguished from mass times acceleration, and so it can be hidden in the mass,

$$m = m_{\rm bare} + \frac{4U_0}{3c^2}. \tag{4}$$

This is the first appearance in the subject of what would later be called mass renormalization, and the source of the first defect below.

The second row is the one that matters, and its coefficient cannot depend on how the charge is arranged. With no power of R available there is nothing left to build a coefficient from except e and c, so the combination is fixed up to a pure number. A sphere, a shell and a dumbbell all return $2e^2/3c^3$, while each returns its own coefficient for the divergent term. That asymmetry is the reason the radiation-reaction term is believed and the mass term is renormalized away.

The third row is the casualty. Those terms are the only place where the internal structure of the electron would ever have shown itself, and the point limit destroys precisely that information.

Substituting Eq. (3) into $m_{\rm bare}\dot{\mathbf{v}} = \mathbf{F} + \mathbf{F}_{\rm self}$ and using Eq. (4) gives

$$\boxed{m\dot{\mathbf{v}} - \frac{2e^2}{3c^3}\ddot{\mathbf{v}} + \cdots = \mathbf{F}} \tag{AL}$$

now usually called the Abraham–Lorentz equation, in which m is the observed mass and everything from the second term on the left is the self-force.

1.3.2. *The same result from energy balance*

There is a much shorter route, which avoids the charge distribution altogether and therefore never meets the divergent mass. Take the Larmor formula for the radiated power, $P = (2e^2/3c^3)\,\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}$, and demand that the work done by the self-force account for the energy radiated over an interval,

$$\int_{t_1}^{t_2} \mathbf{F}_{\text{self}} \cdot \mathbf{v}\, dt = -\int_{t_1}^{t_2} \frac{2e^2}{3c^3}\, \dot{\mathbf{v}} \cdot \dot{\mathbf{v}}\, dt. \tag{5}$$

Integrating the right-hand side by parts,

$$-\int_{t_1}^{t_2} \frac{2e^2}{3c^3} \dot{\mathbf{v}} \cdot \dot{\mathbf{v}}\, dt = -\frac{2e^2}{3c^3} \Big[\dot{\mathbf{v}} \cdot \mathbf{v}\Big]_{t_1}^{t_2} + \int_{t_1}^{t_2} \frac{2e^2}{3c^3}\, \ddot{\mathbf{v}} \cdot \mathbf{v}\, dt. \tag{6}$$

If the boundary term vanishes — for periodic motion, or whenever $\dot{\mathbf{v}} \cdot \mathbf{v}$ takes the same value at both ends — the two integrands may be compared, and

$$\mathbf{F}_{\text{self}} = \frac{2e^2}{3c^3}\, \ddot{\mathbf{v}}, \tag{7}$$

which is the self-force of Eq. (AL) in three lines. The mass never enters, so it stays where classical mechanics put it.

Two things should be said about what this shortcut costs. First, the conclusion is drawn from equality of integrals over a restricted class of intervals, not from any pointwise identity, so it establishes an average effect and not an instantaneous law; the third and higher derivatives of $\mathbf{v}$, which the direct calculation delivers in Eq. (3), are simply invisible to it. Second, and more interestingly, the term thrown away is not junk. The boundary term $(2e^2/3c^3)\, \dot{\mathbf{v}} \cdot \mathbf{v}$ is the Schott energy, and the piece of the covariant reaction force whose time derivative it is will turn out to be the Schott term. Everything awkward about energy balance in Sec. 3 — where the radiated power and the work done by the reaction force refuse to match instant by instant, and where uniformly accelerated motion radiates while feeling no reaction at all — is already present here, in the term this derivation is allowed to discard because the motion was assumed periodic.

1.3.3. *Three defects*

Three objections can be raised against Eq. (AL).

The mass is entirely electromagnetic, and comes out with the wrong coefficient. The absorbed term gives $m = 4U_0/3c^2$, where U_0 is the electrostatic self-energy, finite here because the radius is. The factor $4/3$ is itself a symptom: a covariant accounting of the momentum of a bound charge distribution yields U_0/c^2, and the extra third appears only because the electromagnetic stress has been counted while whatever holds the sphere together has not — the same omission as before, now showing up as a number. Fermi and Wilson later showed that the coefficient becomes unity once the binding stresses are included. The deeper objection survives the repair. If inertia is electromagnetic in origin, a neutral particle should have no mass, and neutral particles have mass. Inertial mass cannot be a manifestation of charge; at most one may say that charge brings mass with it. Nor does Lorentz's derivation claim otherwise on its own terms, since the U_0 appearing in it is electromagnetic energy alone and contains no mechanical rest energy. The energy-balance derivation of Sec. 1.3.2 sidesteps the whole question, which is its chief attraction.

The equation is not covariant. Equation (AL) is written with d/dt in a particular frame. The repair is due to Landau and Lifshitz and is disarmingly short. Replace d/dt by $d/d\tau$ and guess $m\,\dot{u}^\alpha = F^\alpha + k\,\ddot{u}^\alpha$; the guess fails at once, because contracting with u_α and using $u^\alpha u_\alpha = -1$, whose first two derivatives give $u \cdot a = 0$ and $u \cdot \dot{a} = -a \cdot a$, leaves $-k\,(a \cdot a)$ on the right where zero is required. The cure is to add a term along u^α and let the constraint fix its coefficient: demanding $u_\alpha(\dot{a}^\alpha + C\,u^\alpha a \cdot a) = 0$ gives $C = -1$, so that the reaction force must be

$$\frac{2e^2}{3c^3}\left(\dot{a}^\alpha - u^\alpha a^\beta a_\beta\right) = \frac{2e^2}{3c^3}\left(\delta^\alpha{}_\beta + u^\alpha u_\beta\right)\dot{a}^\beta. \tag{8}$$

This is exactly the term that Dirac's far longer construction produces in Sec. 2. Covariance alone, applied to a nonrelativistic result, very nearly hands over the answer — which is worth keeping in mind when judging how much the tube derivation really establishes.

A free electron accelerates itself without bound. This is the serious defect, and it is not repaired anywhere in these notes. It is the subject of Sec. 1.4.

1.4. *The Abraham–Lorentz equation and its runaway solutions*

Set $\mathbf{F} = 0$ in Eq. (AL), keep only the two terms written, and reduce to one dimension. The coefficient of the self-force term is m times a time,

$$\tau_0 = \frac{2e^2}{3mc^3} = 6.27 \times 10^{-24}\ \mathrm{s} \quad \text{(electron)}, \tag{9}$$

which will reappear in every section below, and the equation of motion of a free electron collapses to a first-order equation for the acceleration alone,

$$\dot{a} = \frac{a}{\tau_0}, \qquad \text{whence} \qquad a = C\,e^{t/\tau_0}. \tag{10}$$

Both branches of this solution are unsatisfactory, and for different reasons. If $C \neq 0$, an electron with nothing whatever acting on it accelerates itself, and does so on the timescale τ_0, which is to say it gains a factor of e in acceleration every $\sim 10^{-23}$ seconds. In the nonrelativistic equation the velocity grows without bound; in the covariant version of Sec. 3 it approaches c instead, but the character of the solution is unchanged. No intuition about free particles survives this, and no agent is available to be held responsible for the energy.

If $C = 0$ then $a = 0$, the electron moves uniformly, and Newton's first law is recovered; one may say that the Abraham–Lorentz equation tells us that a free electron moves exactly as an uncharged body does. The physical content is unobjectionable. What is objectionable is the standing of the condition. Equation (10) is first order in a and therefore admits any initial value $a(0)$; the physical solution is picked out by setting $a(0) = 0$, and the only reason offered for doing so is that every other choice misbehaves as $t \to \infty$. A condition imposed at infinity to restrict data at $t = 0$ has no warrant in the initial-value problem it is being applied to, and this remains true no matter how reasonable the surviving solution looks.

The impulse problem sharpens the objection into something concrete. Write the electron's equation with an impulsive force of strength J delivered at $t = 0$,

$$ma - m\tau_0\,\dot{a} = J\,\delta(t), \tag{11}$$

and compare with the corresponding classical problem. A free mass point struck by an impulse jumps discontinuously in velocity and has zero acceleration immediately afterwards: the blow is over, and so is the acceleration. Solving Eq. (11) with the electron moving uniformly beforehand gives instead

$$a = \begin{cases} 0, & t < 0, \\ \kappa\, e^{t/\tau_0}, & t > 0, \end{cases} \qquad \kappa = -\frac{J}{m\tau_0}. \tag{12}$$

The electron does not stop accelerating when the blow ends. It accelerates ever harder, without limit, long after nothing is acting on it, and — since κ carries the opposite sign to J — it runs away in the direction opposite to the blow that started it. This is the runaway, or acceleration collapse, in its starkest form, and no choice of initial data avoids it here: the requirement of uniform motion before the pulse has already been spent.

The root of the behaviour is visible in the order of the equation. Newton's equation of motion contains the acceleration and nothing beyond it, so when the force stops the acceleration stops with it. Equation (AL) contains derivatives of the acceleration, so the state of the electron at a given instant includes how fast its acceleration is changing, and that quantity survives the force that produced it. What sustains it, physically, is the electron's own electromagnetic field. The defect is therefore not an artefact of the sphere model or of the expansion used to reach Eq. (AL); any equation of motion that accounts for the self-field by adding a derivative of the acceleration will inherit it. Dirac's equation, derived on entirely different principles in Sec. 2, inherits it exactly.

1.5. *Criteria for a physically acceptable equation of motion*

Everything from here on consists of proposals to replace Eq. (AL) with something better behaved, and proposals cannot be judged without a standard to judge them by. It is worth fixing that standard now, before any candidate is on the table, so that it cannot be quietly adjusted to fit whichever equation is under discussion. Three requirements are imposed throughout these notes.

1. **The motion must follow from the equation in the ordinary way.** The particle should obey a differential equation that can be solved by the usual procedure — supply initial or boundary data, integrate, obtain the state at all later times — and the data supplied must themselves be data the physical situation could actually provide. This is a weaker demand than it may appear, and Sec. 1.4 has already shown Eq. (AL) failing it: the equation is solvable, but the solution that survives is picked out by a condition imposed at $t \to \infty$, and no experimenter preparing an electron at $t = 0$ has access to that condition.

2. **A stable state must remain stable under a small disturbance.** The concrete test used throughout is the one already applied in Sec. 1.4: take a particle in uniform motion, strike it with a pulse, and ask what it does afterwards. An acceptable equation returns it to uniform motion. It is also acceptable — and this weaker alternative is included deliberately, because Sec. 5 produces solutions of exactly this kind — for the particle to settle into a regular variation confined to a bounded range of speeds. What is not acceptable is for the particle to accelerate without limit toward c, or to swing back and forth between $+c$ and $-c$. Both are read here as signs that the equation has no stable state to return to, rather than as physics.

3. **The ordinary physical laws must hold.** Energy conservation and causality are not negotiable, and an equation that purchases good behaviour by breaking either has not solved the problem but relocated it. This criterion does the least work in the pages that follow and is the

easiest to forget, which is why it is worth writing down: the sign of a radiation-reaction term, for instance, is exactly the kind of thing one may change to remove a runaway without noticing what it does to the energy budget.

These three are the scorecard. Sections 3, 4 and 5 each close by returning to them. Two features of the list are worth noticing before it is used. It is stated entirely in terms of the behaviour of solutions and says nothing about the form an equation should take, which is deliberate, since the strategy of the sections that follow is precisely to alter the form and inspect what comes out. And it asks that a charged particle behave, in these respects, like an ordinary mechanical one — an expectation that is reasonable, widely shared, and worth flagging as an assumption rather than a deduction.

2. THE LORENTZ–DIRAC EQUATION: DERIVATION

[Terminology: throughout these notes "Lorentz–Dirac" (LD, also called Abraham–Lorentz–Dirac) refers to the classical equation of motion of a point charge, never to Dirac's relativistic wave equation.]

2.1. *The world-tube momentum–energy balance*

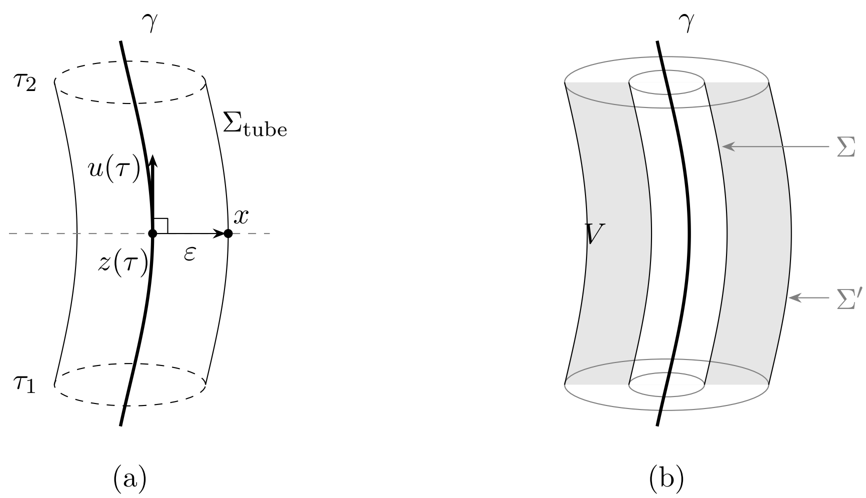

Figure 1. Dirac's construction. *(a)* The tube. At each proper time the point $z(\tau)$ on the world line γ carries a sphere of radius ε, drawn in the hyperplane of simultaneity of $z(\tau)$ — the dashed line — so that the separation $x - z(\tau)$ to a point x of the tube is orthogonal to the four-velocity $u(\tau)$, as in Eq. (15). Dragging that sphere along γ from τ_1 to τ_2 sweeps out Σ_{tube}. The end caps are drawn dashed as a reminder that the integration is carried out over the wall alone; see Sec. 2.5. *(b)* Why the choice of tube does not matter. Two tubes Σ and Σ' enclose the same world line, and the region V between them contains no charge, so the flux of $T^{\mu\nu}$ through the two is the same by Eq. (14). The balance may therefore be read on any surface, including one that keeps well away from the singular world line.

Dirac's advance was a change of question. Lorentz's route, followed in Sec. 1.3, asks what force the charge exerts on itself and answers by summing the retarded action of one charge element on another — a sum that has no meaning for a point charge, since the self-field diverges on the world line. Dirac keeps the point charge and declines to ask about the force at all. He asks instead where the energy and momentum go. Whatever the particle radiates must be paid for out of the particle's own energy and momentum, and that balance can be struck without ever evaluating a field on the world line. The divergent self-field still enters the calculation, but only through differences, and the differences are finite.

2.1.1. *The tube, and why its shape is immaterial*

Surround the world line γ by a three-dimensional tube Σ. The four-momentum crossing it is

$$\Delta P^\mu = \int_\Sigma T^{\mu\nu}\, d\Sigma_\nu. \tag{13}$$

Let Σ and Σ' both enclose γ, and let V be the region between them. That region contains no charge, so the stress–energy tensor is conserved throughout it, and

$$\Delta P'^\mu - \Delta P^\mu = \oint_{\partial V} T^{\mu\nu}\, d\Sigma_\nu = \int_V T^{\mu\nu}{}_{,\nu}\, dV = 0. \tag{14}$$

This is the whole reason the strategy beats Lorentz's, and it deserves to be stated on its own. A force must be evaluated somewhere, and for a point charge every candidate location is a place where the field is infinite. A balance need not be evaluated anywhere in particular. It may be read off on any surface at all, including surfaces that keep a comfortable distance from the singularity, and Eq. (14) guarantees that the reading is the same. See Fig. 1(b).

Dirac's tube is the one adapted to the instantaneous rest frame. For a point x on the tube, let $z(\tau)$ be the point of γ satisfying

$$[x^\mu - z^\mu(\tau)]\, u_\mu(\tau) = 0, \qquad [x - z(\tau)] \cdot [x - z(\tau)] = \varepsilon^2, \tag{15}$$

the separation being spacelike. The first condition places x on the hyperplane of simultaneity of $z(\tau)$, so that "at the same moment" means at the same moment for the particle; the second fixes the radius. Together they describe a sphere of radius ε drawn in the instantaneous rest frame and dragged along the world line, sweeping out $\Sigma_{\mathrm{tube}} = \{x(\tau, \varepsilon, \Omega);\ \tau_1 \le \tau \le \tau_2\}$, as in Fig. 1(a).

2.1.2. *The surface element*

Set $c = 1$ for the rest of this subsection and work in the instantaneous rest frame at proper time τ, restoring c at the end. Let n^μ be the unit spacelike vector pointing from $z(\tau)$ to the point x of the tube, so that $x = z(\tau) + \varepsilon n$ with $n \cdot u = 0$ and $n \cdot n = 1$. Three angular integrals are all that will be needed:

$$\oint d\Omega = 4\pi, \qquad \oint n^i\, d\Omega = 0, \qquad \oint n^i n^j\, d\Omega = \frac{4\pi}{3}\delta^{ij}. \tag{16}$$

Differentiate $x = z + \varepsilon n$ along the world line at fixed Ω. The vector n is Fermi–Walker transported, so $\dot{n}^\mu = (a \cdot n)\, u^\mu$, and

$$\frac{\partial x^\mu}{\partial \tau} = u^\mu \left(1 + \varepsilon\, a \cdot n\right). \tag{17}$$

Each of the two angular directions contributes a factor ε, so

$$d\Sigma_\nu = n_\nu\, \varepsilon^2 \left(1 + \varepsilon\, a \cdot n\right) d\Omega\, d\tau. \tag{18}$$

The factor $1 + \varepsilon\, a \cdot n$ is the one thing in this subsection worth memorizing. It is there because the tube is dragged along a world line that is accelerating: successive cross-sections are not parallel, and the wall stretches on the side the particle is accelerating towards. It is also easy to write the surface element without it, and everything that distinguishes this calculation from the charged sphere of Sec. 1.3 is contained in it.

2.1.3. *The field near the world line*

Expanded about $z(\tau)$ in the rest frame, the retarded field is

$$\mathbf{E} = \frac{e}{\varepsilon^2}\,\mathbf{n} - \frac{e}{2\varepsilon}\left[\mathbf{a} + (\mathbf{a}\cdot\mathbf{n})\mathbf{n}\right] + \frac{2e}{3}\,\dot{\mathbf{a}} + O(\varepsilon), \tag{19}$$

while $\mathbf{B}$, which vanishes for a charge at rest, begins only at $O(\varepsilon^{-1})$. The surface element carries ε^2, so a product of two fields must reach $O(\varepsilon^{-3})$ to leave anything divergent behind. Since $\mathbf{BB}$ is only $O(\varepsilon^{-2})$, the magnetic field cannot contribute to the divergent term at all, and the whole of the next subsubsection is an exercise with $\mathbf{E}$. The four-momentum leaving the tube is

$$\frac{dP^\mu}{d\tau} = -\oint T^{\mu\nu} n_\nu\, \varepsilon^2\,(1 + \varepsilon\, a\cdot n)\, d\Omega, \qquad T^{\mu\nu} = \frac{1}{4\pi}\left[F^{\mu\alpha}F^{\nu}{}_{\alpha} - \tfrac{1}{4}\eta^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}\right]. \tag{20}$$

2.1.4. *The divergent term, and what becomes of the* 4/3

Two products of Eq. (19) reach the required order. Take the Coulomb term against itself first:

$$T^{ij}n_j\Big|_{\mathrm{CC}} = \frac{1}{4\pi}\frac{e^2}{\varepsilon^4}\left[n^i n^j - \tfrac{1}{2}\delta^{ij}\right] n^j = \frac{e^2}{8\pi\varepsilon^4}\, n^i. \tag{21}$$

At order ε^{-2} this integrates to nothing, since $\oint n^i d\Omega = 0$: an isotropic Coulomb field pushes equally in every direction. Against the tilt factor it survives,

$$-\oint \frac{e^2}{8\pi\varepsilon^4}\, n^i\, \varepsilon^3 (a\cdot n)\, d\Omega = -\frac{e^2}{8\pi\varepsilon}\cdot\frac{4\pi}{3}a^i = -\frac{e^2}{6\varepsilon}\, a^i. \tag{22}$$

The second product is the Coulomb term against the acceleration field, which is the one a sphere would have too:

$$-\oint \frac{1}{4\pi}\left[-\frac{e^2}{2\varepsilon^3}a^i - \frac{e^2}{2\varepsilon^3}(a\cdot n)n^i\right]\varepsilon^2 d\Omega = \frac{e^2}{2\varepsilon}\left(1 + \tfrac{1}{3}\right)a^i = \frac{2e^2}{3\varepsilon}\, a^i. \tag{23}$$

Writing $U_0 = e^2/2\varepsilon$ for the electrostatic energy of a shell of radius ε, the coefficient here is $4U_0/3$: the same 4/3 the sphere produced in Sec. 1.3, arrived at in the same way, by ignoring the tilt.

Adding Eqs. (22) and (23),

$$\frac{dP^i}{d\tau}\Big|_{\mathrm{div}} = \left(\frac{2}{3} - \frac{1}{6}\right)\frac{e^2}{\varepsilon}\, a^i = \frac{e^2}{2\varepsilon}\, a^i = U_0\, a^i. \tag{24}$$

The coefficient is 1, not 4/3, and the two derivations in these notes therefore disagree about the electromagnetic mass: Eq. (4) absorbed $4U_0/3c^2$, while Eq. (24) absorbs U_0/c^2. Neither contains an error, and it is worth being clear about what the disagreement is.

Section 1.3 recorded the 4/3 as the mark of having counted the electromagnetic stress while omitting whatever holds the sphere together. Dirac's calculation has no sphere to hold together and no binding stress to omit, and it returns U_0/c^2. What produces the difference is the tilt factor of Eq. (18), and the tilt factor is a statement about simultaneity: Lorentz gathers the field momentum on a hyperplane of constant time in one chosen frame, Dirac gathers it on a surface assembled from the particle's own succession of rest frames. The gap between those two ways of deciding what counts as "at the same moment" is precisely the missing third.

The 4/3 was never a statement about electrons. It was a statement about a choice of simultaneity, and it disappears as soon as the choice is made covariantly.

2.1.5. *The finite term, and two checks*

The cross of the Coulomb field with the radiative term $\frac{2e}{3}\dot{\mathbf{a}}$ gives, after the first and third pieces of $T^{ij}n_j$ cancel against each other,

$$-\oint \frac{1}{4\pi}\frac{2e^2}{3\varepsilon^2}\,\dot{a}^i\,\varepsilon^2\,d\Omega = -\frac{2e^2}{3}\,\dot{a}^i. \tag{25}$$

The finite pieces not written out — the acceleration field against itself, $\mathbf{B}$ against itself, and the tilt correction to Eq. (23) — contain no $\dot{a}$, and assemble into the term along u^μ that $u\cdot dP/d\tau = 0$ requires. Restoring c,

$$\frac{dP^\mu}{d\tau} = \frac{e^2}{2\varepsilon c^2}\,a^\mu - \frac{2e^2}{3c^3}\left(\dot{a}^\mu - u^\mu a^\beta a_\beta\right), \tag{26}$$

P^μ being the four-momentum carried out of the tube.

Two things can be checked without redoing anything. Contracting Eq. (26) with u_μ, and using $u\cdot u = -1$ and $u\cdot\dot{a} = -a\cdot a$, gives zero, as it must. And the component along u^μ is $(2e^2/3c^3)\,a\cdot a$, the Larmor power — so the radiated energy is present with the coefficient that Sec. 1.3.2 obtained from energy balance, by an argument that never mentioned a tube.

What Eq. (26) is not is an equation of motion. It records how much four-momentum the *field* carries out of the tube. Turning that into a statement about the particle requires saying what the particle's own four-momentum is, and nothing in the calculation supplies it. That freedom is the subject of Sec. 2.2.

2.2. *The undetermined four-vector B^μ*

Total four-momentum is conserved, so the particle's own four-momentum changes at the rate at which the external field supplies it, less the rate at which the field carries it out of the tube,

$$\frac{dP^\mu_{\text{mech}}}{d\tau} = F^\mu_{\text{ext}} - \frac{dP^\mu}{d\tau}, \tag{27}$$

with $dP^\mu/d\tau$ given by Eq. (26). Everything on the right is known. The left is not, because nothing computed so far says what P^μ_{mech} is.

This is the bill for the trick of Sec. 2.1.1. The tube integral was arranged to be insensitive to what lies inside it — that insensitivity is exactly why the divergent self-field never had to be evaluated — and the particle lies inside it. The construction is silent about the interior by design, and it is silent about the particle's own momentum for the same reason. So a form has to be put in by hand:

$$P^\mu_{\text{mech}} = m_0 u^\mu + B^\mu, \tag{28}$$

with B^μ undetermined.

One condition is available. Since $u\cdot u = -1$ throughout, $u\cdot a = 0$, and contracting Eq. (27) with u_μ must therefore give an identity. The Lorentz force is orthogonal to u, and $u\cdot dP/d\tau = 0$ was checked in Sec. 2.1.5, and $m_0\,u\cdot a = 0$; so what remains is

$$u_\mu\,\frac{dB^\mu}{d\tau} = 0. \tag{29}$$

This says no more than that whatever force B^μ contributes must be orthogonal to the four-velocity, which is a requirement on any force term in a relativistic equation of motion. It is the only condition the construction imposes.

It is a very weak condition: one scalar equation restricting four functions, so three functions of τ remain free at every instant. Dirac's choice is the simplest solution,

$$B^\mu = k\, u^\mu, \qquad k = \text{const}, \tag{30}$$

for which $dB^\mu/d\tau = k\, a^\mu$ and Eq. (29) holds automatically. Substituting Eqs. (28) and (30) into Eq. (27) with Eq. (26),

$$\left(m_0 + k + \frac{e^2}{2\varepsilon c^2}\right) a^\mu = F^\mu_{\text{ext}} + \frac{2e^2}{3c^3}\left(\dot{a}^\mu - u^\mu a^\beta a_\beta\right), \tag{31}$$

and the entire content of Dirac's choice is visible in the bracket: k appears only added to m_0 and to the divergent $e^2/2\varepsilon c^2$. Choosing $B^\mu = ku^\mu$ is choosing to add nothing to the dynamics. It shifts a mass, and the mass is not observable separately in any case, since only the sum

$$m = m_0 + k + \frac{e^2}{2\varepsilon c^2} \tag{32}$$

appears. Holding m finite as $\varepsilon \to 0$ requires $m_0 + k \to -\infty$, which should be recorded plainly: the bare mass is negatively infinite, and the observed mass is the residue of a cancellation between two quantities neither of which is separately meaningful.

Two things follow, and they point in opposite directions.

The first is that the derivation has not determined the equation of motion. It has determined it *given* Eq. (30), and Eq. (30) was chosen for simplicity, not derived. Any

$$B^\mu = P_0\, u^\mu + P_1\, a^\mu + P_2\, \dot{a}^\mu + \cdots \tag{33}$$

whose derivative is orthogonal to u is equally admissible, and each choice yields a different equation of motion — generally of higher order, since each extra term carries another derivative. Dirac himself put it as a choice of the simplest form rather than as a result. Section 5 takes that door seriously and walks through it.

The second is that the freedom is less mysterious than the notation makes it look, because a particular choice of B^μ is nothing but a choice of what to call the particle's momentum. Suppose one posits, instead of Eq. (30),

$$P^\mu_{\text{mech}} = m_0 u^\mu + Ce^2 a^\mu \tag{34}$$

and asks what C must be. Contracting the source-free balance with u_μ: the mechanical side gives $u \cdot (m_0 a + Ce^2 \dot{a}) = -Ce^2\, a \cdot a$, the field side gives $-\frac{2}{3}e^2 a \cdot a$ from the radiated part of Eq. (26), and the two cancel only if

$$C = -\tfrac{2}{3}. \tag{35}$$

So orthogonality does fix the coefficient, once the *form* has been assumed — which is the same situation as before, wearing different clothes. What is worth noticing is the object it fixes. The extra piece of the momentum is $-\frac{2}{3}e^2 a^\mu$, which is the Schott momentum: the same quantity that appeared in Sec. 1.3.2 as the boundary term $(2e^2/3c^3)\, \dot{\mathbf{v}} \cdot \mathbf{v}$ that the energy-balance derivation was permitted to discard. It was discarded there by assuming periodic motion. Here it cannot be discarded, and it has to be carried as part of what the particle owns.

2.3. *The equation*

Absorbing the mass as in Eq. (32), Eq. (31) is the Lorentz–Dirac equation,

$$\boxed{ma^\alpha = F^\alpha_{\text{ext}} + \frac{2e^2}{3c^3}\left(\delta^\alpha{}_\beta + u^\alpha u_\beta\right)\dot{a}^\beta} \tag{LD}$$

the projector $\delta^\alpha{}_\beta + u^\alpha u_\beta$ being the compact way of writing $\dot{a}^\alpha - u^\alpha a\cdot a$, and enforcing $u\cdot a = 0$ term by term.

The reaction force has two pieces and they are not the same kind of thing. The piece $\frac{2e^2}{3c^3}\dot{a}^\alpha$ is the Schott term. It is a total derivative, it changes sign with $\dot{a}$, and it is not radiation: it is momentum lent to the near field and returned. The piece $-\frac{2e^2}{3c^3}u^\alpha a\cdot a$ points along the four-velocity, never changes sign, and is exactly the Larmor power carried off, as Sec. 2.1.5 verified. Only the second is irreversible. Almost every puzzle in Sec. 3 — the failure of the radiated power and the work done by the reaction force to match instant by instant, and uniform acceleration radiating while feeling no reaction at all — is a puzzle about the first.

It is worth stating what this derivation did and did not buy, since the same equation was very nearly written down in Sec. 1.3 by Landau and Lifshitz's two-line covariantization, Eq. (8). What is new is not the form. Covariance and $u\cdot a = 0$ fix the form almost by themselves, and the coefficient $2e^2/3c^3$ was already fixed by the nonrelativistic limit. What is new is, first, that no extended body was needed and therefore no binding stress was omitted, which is why the electromagnetic mass came out as U_0/c^2 rather than $4U_0/3c^2$; and second, that the freedom left over has been made explicit rather than hidden, in the shape of Eq. (33). A derivation that tells you exactly which of its conclusions were choices is worth more than one that reaches the same equation silently.

What the derivation did not buy is any improvement in the behaviour of the solutions. Equation (LD) still contains $\dot{a}$, so it is still of third order in the position, and Sec. 1.4 already showed what that implies. Section 3 works out the consequences.

2.4. *The other route: separating the singular part*

There is a second way through, and it has the opposite character. Dirac's route declines to ask what force the charge exerts on itself and audits the momentum instead. This one asks about the force after all, but subtracts, before asking, the part of the self-field that makes the question meaningless.

The difficulty being addressed is the one stated in Sec. 1: at the electron both the potential and the field tensor are infinite, so the Lorentz force expression cannot be applied to the self-field. The proposal is to split the infinite self-field into an "irrelevant" singular part, which is discarded, and a finite "effective" part, and then to apply the Lorentz force law to the effective part alone, in the ordinary way. The discarded piece is called the singular term, and the procedure is the method of separating the singular part.

The split that works is between the advanced and retarded potentials. The advanced potential A_{adv} violates causality and is not a candidate for the physical field, but it is a perfectly good solution of $\Box A^\alpha = -4\pi j^\alpha$, and near the world line its singular behaviour is identical to that of A_{ret}. In the difference the singular behaviour cancels. What is kept and what is thrown away are therefore

$$A^\alpha_{(R)} = \tfrac{1}{2}\left(A^\alpha_{\text{ret}} - A^\alpha_{\text{adv}}\right), \qquad A^\alpha_{(S)} = \tfrac{1}{2}\left(A^\alpha_{\text{ret}} + A^\alpha_{\text{adv}}\right), \tag{36}$$

the first being a difference of two solutions of the inhomogeneous equation and hence a solution of the homogeneous Maxwell equations, regular on the world line, with a finite Lorentz force $f^\alpha = e\,F^{(R)\alpha}{}_\beta u^\beta$.

That this reproduces Dirac's answer is the content of the equivalence theorem:

> The motion obtained from momentum–energy conservation is the same as the motion produced by the Lorentz force of the potential $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$.

Part of the check is already in hand. The finite term computed in Sec. 2.1.5 is precisely e times $\frac{1}{2}(F_{\rm ret} - F_{\rm adv})$ evaluated on the world line, which is why the same $2e^2/3c^3$ appeared there. The agreement of the two routes is not an accident, but neither is it obvious: one of them never evaluates a field at the particle and the other does nothing else.

The proof is deferred to Sec. 4.1, and it is worth saying why it gets a section of its own rather than a footnote. Its value is not that it confirms Eq. (LD) a third time. It is that carrying the proof out makes visible something the statement conceals: that Eq. (36) is a *choice* of which combination to discard, and that the choice was made on grounds of convenience. Section 4 relaxes it, replacing Eq. (36) by a two-parameter family, and everything in that section descends from the observation that the split is not forced.

2.5. *The end caps, and where the freedom in B^μ comes from*

One reservation about the derivation remains to be entered, and it turns out not to be independent of the freedom found in Sec. 2.2.

The region V occupied by the tube between τ_1 and τ_2 is bounded by the wall $\Sigma_{\rm tube}$ and by two spacelike end caps $C(\tau_1)$ and $C(\tau_2)$, the discs closing the tube at each end. Conservation applies to the whole boundary, not to part of it:

$$\int_{C(\tau_2)} T^{\mu\nu} d\Sigma_\nu - \int_{C(\tau_1)} T^{\mu\nu} d\Sigma_\nu + \int_{\Sigma_{\rm tube}} T^{\mu\nu} d\Sigma_\nu = \int_V T^{\mu\nu}{}_{,\nu}\, dV. \tag{37}$$

Both the calculation of Sec. 2.1 and Poisson's shorter version evaluate the third term only. The 2006 text records this and does not pursue it.

The two cap integrals are not incidental. Each is the four-momentum of the field contained inside the tube at one end,

$$P^\mu_{\rm field}(\tau) = \int_{C(\tau)} T^{\mu\nu} d\Sigma_\nu, \tag{38}$$

so what Eq. (37) says is that the wall integral measures the change in the field momentum *inside* the tube, and not the change in the particle's momentum. Those are different quantities, and separating them requires knowing how the contents of the tube divide between the particle and its field.

Which is exactly what Sec. 2.2 had to put in by hand. If Eq. (38) could be evaluated, $P^\mu_{\rm mech}$ would follow from it rather than being posited, Eq. (28) would not be needed, and the four-vector B^μ would have no freedom left in it. The arbitrariness in B^μ is not a second defect standing alongside the missing caps. It is the missing caps, seen from the side of the equation of motion.

The caps are omitted because they diverge. On a cap the Coulomb energy density goes as e^2/r^4 against a volume element $r^2 dr$, and the momentum density, carrying one factor of $\mathbf{B} \sim 1/r$, as e^2/r^3; the two radial integrals

$$\int_0 \frac{dr}{r^2}, \qquad \int_0 \frac{dr}{r} \tag{39}$$

diverge at the lower limit, the second only logarithmically but no less fatally. The field momentum inside the tube is infinite, which is why it cannot be apportioned, which is why a form for $P^{\mu}_{\rm mech}$ has to be assumed.

This closes the account of Sec. 2.1 in a way worth stating plainly. The virtue of the tube construction and its cost are the same property. It works because it is insensitive to what lies inside it, and it is for that reason unable to say anything about what lies inside it. The divergence was not removed. It was moved to a place where it could be ignored, and the price of ignoring it is one undetermined four-vector.

A further consequence of the same divergence is left aside here. The stress–energy tensor of the electron's own field curves spacetime, and a quantity that diverges on the world line curves it without bound, so the particle is not in Minkowski space and general relativity has nothing to say about the region where the problem lives. The 2006 text raises this in its Sec. 4 and builds an "effective metric" proposal on it; that material is not covered in these notes.

3. THE LORENTZ–DIRAC EQUATION: DISCUSSION AND SOLUTIONS

3.1. *One-dimensional reduction*

[Parametrize by the rapidity q via $\dot{t} = \cosh q$, $\dot{x} = \sinh q$. Equation (LD) collapses to a single scalar equation:]

$$\dot{q} - \tfrac{2}{3}\ddot{q} = F^{01}. \tag{40}$$

3.2. *The free solution and the runaway*

$$q = C_1 e^{3\tau/2} + C_2. \tag{41}$$

[$C_1 \neq 0$ is the runaway; $C_1 = 0$ has to be imposed by hand. Note that no initial condition on position and velocity alone selects it — the equation is third order.]

3.3. *Dirac's asymptotic condition and the price paid*

[Pushing the condition out to $\tau \to \pm\infty$ eliminates the runaway but introduces pre-acceleration: the particle starts moving before the force arrives, on a timescale τ_0. This trades criterion (2) for criterion (3) of Sec. 1.5.]

3.4. *Energy balance and the Schott term; hyperbolic motion*

[Where the energy sits during acceleration, and the standing puzzle of uniformly accelerated motion, in which the reaction term vanishes although the particle radiates.]

3.5. *Scorecard, and the fork in the road*

[Against the three criteria of Sec. 1.5, LD fails all three in one form or another. Two ways to modify it then present themselves, and they are the subjects of the next two sections:

- sideways — *keep the structure of the reaction term and open up its coefficient (Sec. 4);*
- upward — *keep the coefficient and raise the order of the equation by enlarging B^{μ} (Sec. 5).*

This contrast is the spine of the notes.]

4. SPLITTING OFF THE SINGULAR PART: THE λ–κ FAMILY

4.1. *The equivalence theorem*

[The motion obtained from momentum–energy conservation is the same as the motion obtained by applying the Lorentz force law with $\frac{1}{2}(A_{\mathrm{ret}} - A_{\mathrm{adv}})$. Give the elementary proof from the 2006 text (Taylor expansion of the retarded potential, term-by-term cancellation), which is more direct than the routes via Zhang Zongsui or Poisson.]

4.2. *Does the singular part act?*

[Construct the stress–energy tensor of $A_S = \frac{1}{2}(A_{\mathrm{ret}} + A_{\mathrm{adv}})$ and integrate it. Eq. (1.2.17) of the 2006 text:]

$$\frac{dP^\alpha}{d\tau} = \frac{a^\alpha}{2r} + \frac{2}{3}\dot{a}^\alpha. \tag{42}$$

[The 2006 text read the surviving second term as showing that the singular part is not *inert, and used that as the licence to loosen the split. Flag: this is the load-bearing step of the whole section, and it needs to be checked — see the closing remark of Sec. 4.7.]*

4.3. *The two-parameter family*

$$A^\alpha = \lambda A^\alpha_{\mathrm{ret}} + \kappa A^\alpha_{\mathrm{adv}}. \tag{43}$$

[Two branches are then singled out by requiring that the retained part have a sensible field-theoretic status.]

4.4. *Branch (a): $\lambda + \kappa = 0$*

[The effective part satisfies the homogeneous Maxwell equations. The free-particle a–v phase portrait has a closed-form solution:]

$$a = \left[-\frac{3}{8\kappa}\ln\frac{1+v}{1-v} + C\right](1-v^2)^{3/2}. \tag{44}$$

[$\kappa = -1/2$ recovers LD. The resulting equation of motion is Eq. (1.5.1) of the 2006 text:]

$$ma^\alpha = F^\alpha_{\mathrm{ext}} - \frac{4\kappa}{3}e^2\left(\dot{a}^\alpha - a^2u^\alpha\right). \tag{45}$$

[Behaviour: $\kappa < 0$ gives a convex curve and still runs away; $\kappa > 0$ gives a concave curve that must cross the v axis, so a perturbed particle decelerates *back to uniform motion. The 2006 text called this "Aristotle motion," since force then fixes velocity rather than acceleration.]*

4.5. *Branch (b): $1 - \lambda = -\kappa$*

[Retarded and advanced parts enter the discarded singular term on equal footing. Solve numerically in κ. Result: $\kappa = -1$ is the watershed. For $\kappa > -1$ the runaway survives; for $\kappa < -1$ a critical velocity v_c appears — a particle starting below v_c stays below it, one starting above decelerates stably. Reproduce the table:]

κ	-1.5	-2	-3	-500	-5000
v_c	0.3473	0.2261	0.1341	6.67×10^{-3}	6.67×10^{-4}

4.6. *An unresolved point*

[Extending branch (b) beyond one dimension runs into $\mathbf{n}\cdot\mathbf{v}$ *being undefined at the position of the charge itself. The 2006 text suggested the replacement* $\mathbf{n}\cdot\mathbf{v}\to\frac{1}{3}v$ *but did not carry it out.]*

4.7. *What this branch amounts to*

[Eq. (45) is nothing but the LD equation with the radiation-reaction coefficient $2e^2/3$ *replaced by a free parameter* $-4\kappa e^2/3$*. State this plainly — it is the one result in these notes that outlives its original motivation, whatever becomes of Eq. (42).]*

5. THE ELIEZER-TYPE SERIES

5.1. *A note on the name*

[Placed first, not in a footnote. The equations of this section follow an ansatz for B^μ *taken from Eliezer's 1947 review. They are* not *the equation now standardly called the Eliezer, or Eliezer–Ford–O'Connell, equation, which is obtained by order reduction, is free of runaways, and is a live candidate in present-day experiments. Confusing the two inverts the conclusion of this section.]*

5.2. *Expanding* B^μ

$$B_\mu = P_0 v_\mu + P_1\dot{v}_\mu + P_2\ddot{v}_\mu + \cdots + P_i v_\mu^{(i)} + \cdots \tag{46}$$

[Differentiating $v^\mu v_\mu = -1$ *repeatedly generates a chain of identities whose coefficients form Pascal's triangle; the general term is in Appendix A.4 of the 2006 text. Substituting gives the generalized equation of motion, Eq. (1.3.3) there. Setting all* $P_i = 0$ *returns LD, Eq. (LD).]*

5.3. *Keeping* P_2*: a third-order equation*

$$\dddot{q} - B\ddot{q} + A\dot{q} - \tfrac{1}{2}\dot{q}^3 = 0, \qquad A = \frac{3m}{2k}, \quad B = \frac{e^2}{k}. \tag{47}$$

[Scan the (A, B) *plane numerically. Reported outcome: for* A *and* B *of either common sign the particle accelerates to c on its own; for* $A, B > 0$ *with* $A \gg B$ *the behaviour is worse still, the velocity swinging at high frequency between* $+c$ *and* $-c$*. Figures 1-3-1 through 1-3-4 of the 2006 text.]*

5.4. *Keeping* P_2 *and* P_4*: a fifth-order equation*

[Eq. (1.3.11) of the 2006 text, scanned over 22 parameter triples (A, B, C)*. Three classes of behaviour: oscillation between* $\pm c$*; monotonic approach to c; and a third class, neither reaching c nor returning to uniform motion, in which the speed pulses periodically within a bounded range — "much like an oscillator, except that the oscillator itself keeps moving forward."]*

5.5. *Scorecard*

[No parameter choice in either the third- or the fifth-order family satisfies the criteria of Sec. 1.5. Close with the question the 2006 text closed on — whether nature would really select a more complicated B^μ *— and leave it open here.]*